

APPIUM MOBILE AUTOMATION TEST SUITE: OTP LOGIN

Application: Mobile Banking | Platforms: Android & iOS | Engine: Appium (UiAutomator2 / XCUITest)

TC ID	Priority	Test Category	Description
TC_MOB_OTP_001	P0	Positive / Auth Login	Verify successful user login using a valid registered mobile number and correct 4-digit OTP within 1 minute
TC_MOB_OTP_002	P0	Negative / Invalid OTP	Verify appropriate validation error is displayed when entering an incorrect 4-digit OTP
TC_MOB_OTP_003	P0	OTP Expiry	Verify that an expired OTP (submitted after 1 minute) cannot be used for authentication
TC_MOB_OTP_004	P1	OTP Resend Behavior	Verify Resend OTP button disabled state during 20s cooldown and functional activation after 20 seconds

TC_MOB_OTP_00 5	P0	Max Retry Attempts	Verify account temporary lockout / restriction after 3 consecutive failed OTP submission attempts
TC_MOB_OTP_00 6	P1	Boundary Validation	Verify keypad restrictions and field limits when entering partial (<4 digits) or excess (>4 digits) OTP characters
TC_MOB_OTP_00 7	P1	Boundary / Non-Numeric	Verify input field rejects non-numeric characters, special characters, and white spaces
TC_MOB_OTP_00 8	P1	Network Failure / No Network Request	Verify app behavior when requesting OTP under 'No Network Connection' (Airplane Mode / Offline)
TC_MOB_OTP_00 9	P1	Network Failure / Interrupt Request	Verify app handles network drop during 'Send OTP' API call execution

TC_MOB_OTP_01 0	P1	Network Failure / Interrupt Submission	Verify app resilience and error handling when network disconnects during OTP 'Submit' verification
TC_MOB_OTP_01 1	P2	Slow Network / High Latency	Verify UI loader spinner and request timeout handling under high latency (Slow 2G/3G network profile)
TC_MOB_OTP_01 2	P2	Background App State	Verify OTP screen state, cooldown timer, and input persistence when app is pushed to background and resumed
TC_MOB_OTP_01 3	P2	Platform UI / OS Auto-fill	Verify native Android SMS Retriever API & iOS Keyboard OTP Auto-fill functionality across platforms

Preconditions	Test Steps	Test Data
1. App Installed on Android/iOS device. 2. Device connected via Appium session driver. 3. Mobile number +12015550123 is registered and active.	1. Launch Mobile Banking App. 2. Navigate to Login screen. 3. Enter valid mobile number into `mobile_number_input` field. 4. Tap 'Send OTP' button (`send_otp_btn`). 5. Retrieve 4-digit OTP via SMS gateway API/Mock. 6. Enter 4-digit OTP into	Mobile Number: +12015550123 Valid OTP: 4820 Device: Android 14 / iOS 17
1. App launched on device via Appium. 2. OTP generated for valid mobile number.	1. Enter registered mobile number (+12015550123) and tap 'Send OTP'. 2. Wait for OTP screen to render. 3. Enter incorrect 4-digit code (e.g., 9999). 4. Tap 'Submit' button (`submit_otp_btn`).	Mobile Number: +12015550123 Incorrect OTP: 9999
1. App launched on device. 2. OTP request triggered.	1. Request OTP for valid mobile number (+12015550123). 2. Using Appium wait / system sleep, pause execution for 61 seconds (exceeding 1 min validity). 3. Enter the received 4-digit OTP. 4. Tap 'Submit' button.	Mobile Number: +12015550123 Generated OTP: 4820 Wait Time: 61 seconds
1. OTP screen active in mobile app.	1. Request OTP for valid mobile number. 2. Verify state of 'Resend OTP' button (`resend_otp_btn`) immediately (0 to 19s). 3. Wait 20 seconds for cooldown timer to hit 0. 4. Verify button state after 20 seconds. 5. Tap 'Resend OTP' and fetch new	Mobile Number: +12015550123 Cooldown: 20 seconds

1. App on OTP entry screen via Appium.	1. Enter invalid OTP '0000' and submit (Attempt 1). 2. Enter invalid OTP '1111' and submit (Attempt 2). 3. Enter invalid OTP '2222' and submit (Attempt 3).	Mobile Number: +12015550123 Invalid OTPs: 0000, 1111, 2222
1. Active OTP screen.	1. Tap OTP input field ('otp_input_field'). 2. Enter 3 digits ('123') and check 'Submit' button state. 3. Attempt to enter 5 digits ('12345') via Appium 'send_keys()'.	Partial OTP: 123 Excess OTP: 12345
1. OTP screen active.	1. Focus on 'otp_input_field'. 2. Attempt entering alphabetic strings ('ABCD'), special characters ('@#\$&'), or spaces via Appium driver.	Input Data: 'A@1 '
1. Mobile app open on login screen. 2. Network connection disabled via Appium ('driver.set_network_connection(1)' / Network mock).	1. Disable Wi-Fi and Cellular Data on test device. 2. Enter valid mobile number (+12015550123). 3. Tap 'Send OTP' button.	Mobile Number: +12015550123 Network State: OFF
1. App on mobile login screen with connected network.	1. Tap 'Send OTP'. 2. Interrupt/kill network connection instantly during flight of request (latency simulated via Appium/Proxy).	Mobile Number: +12015550123 Network Toggle: ON -> OFF during request

1. Valid 4-digit OTP entered into input field.	1. Cut off network connection via Appium driver command. 2. Tap 'Submit' button.	OTP: 4820 Network State: Interrupted during submission
1. Appium network throttling set to 'Slow 3G' / 2000ms latency.	1. Enter mobile number and request OTP. 2. Enter valid 4-digit OTP and tap 'Submit'.	Network Profile: 3G / High Latency OTP: 4820
1. App on OTP entry screen with active 20s cooldown timer.	1. Move app to background for 10 seconds via Appium (<code>driver.background_app(10)</code>). 2. Resume app to foreground. 3. Check input field state and cooldown timer integrity.	Background Duration: 10 seconds
1. Real Android device (SMS Retriever) and iOS device (iOS 12+ Keyboard autofill enabled). 2. Active SIM / SMS receiver configured.	1. Trigger OTP to active device. 2. Observe native OS keyboard/SMS popup prompt. 3. Tap native OTP auto-fill suggestion.	Platform: Android 14 / iOS 17 Auto-fill trigger: Active SMS receipt

Expected Result (Appium UI & System State)

1. User is successfully authenticated.
2. Screen redirects to Mobile Banking Dashboard (`dashboard\_view`).
3. Auth Bearer token is saved in App Secure Storage.

1. Error banner/toast displayed: 'Invalid OTP. Please try again.' (`error\_message\_label`).
2. User remains on OTP input screen.
3. Retry counter increments to 1/3.

1. Error message displayed: 'OTP has expired. Please request a new OTP.'
2. Submit action is rejected.
3. User session is not created.

1. 'Resend OTP' button is disabled/greyed out with countdown timer showing (20s -> 0s).
2. At 20s mark, button becomes ENABLED.
3. Tapping 'Resend OTP' dispatches a new 4-digit OTP.

1. On 3rd attempt, explicit modal/alert appears: 'Maximum retry attempts reached. Please request a new OTP or try later.'

2. OTP input field is disabled (`enabled = false`).

3. User is forced to restart OTP flow or wait out restriction.

1. 'Submit' button remains disabled or tapping displays 'Please enter complete 4-digit OTP'.

2. Input field strictly limits entry to max 4 digits ('1234'); 5th digit is ignored/truncated.

1. Mobile soft numeric keypad appears automatically.

2. Field masks/rejects non-numeric characters.

3. Only integer characters (0-9) are accepted into the field.

1. Immediate client-side alert/toast: 'No internet connection. Please check your network and try again.'

2. App does not crash or hang indefinitely.

1. App displays network error banner:

'Connection timeout / lost. Please try again.'

2. Loading spinner terminates gracefully.

1. Friendly error dialog displayed: 'Network error occurred while authenticating. Please try again.'
2. Entered OTP remains in field without clearing or crashing app.

1. Activity loading indicator (`progress\_bar`) displays continuously without UI freeze.
2. Request completes successfully or gracefully times out with user retry prompt after 15s.

1. App resumes seamlessly without crashing.
2. Cooldown timer accurately reflects elapsed background time (reduced by ~10s).
3. Any previously typed OTP digits remain intact.

1. Android SMS prompt / iOS keyboard suggestion bar displays received 4-digit code.
2. Tapping auto-fills all 4 fields seamlessly.
3. Appium locates filled values in `otp\_input\_field`.